

Dynamic hosting capacity analysis for distributed photovoltaic resources—Framework and case study

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ABSTRACT

Distributed photovoltaic systems can cause adverse distribution system impacts, including voltage violations at customer locations and thermal overload of lines, transformers, and other equipment resulting from high current. The installed capacity at which violations first occur and above which would require system upgrades is called the hosting capacity. Current static methods for determining hosting capacity tend to either consider infrequent worst-case snapshots in time and/or capture coarse time and spatial resolution. Because the duration of violations cannot be captured with these traditional methods, the metric thresholds used in these studies conservatively use the strictest constraints given in operating standards, even though both worse voltage performance and higher overloads may be temporarily acceptable. However, assessing the full details requires accurately capturing time-dependence, voltage-regulating equipment operations, and performance of advanced controls-based mitigation techniques. In this paper, we propose a dynamic distributed photovoltaic hosting capacity methodology to address these issues by conducting power flow analysis for a full year. A key contribution is the formulation of time aware metrics to take these annual results and identify the hosting capacity. Through a case study, we show that this approach can more fully capture grid impacts of distributed photovoltaic than traditional methods and the dynamic hosting capacity was 60%–200% higher than the static hosting capacity in this case study.

1. Introduction

The increasing adoption of distributed energy resources (DERs), especially distributed photovoltaics (DPV), and the associated potential for adverse grid impacts has created the need for DER impact studies [1]. Of particular interest is a feeder's DPV hosting capacity—the installed DPV capacity at which violations first occur and above which would require system upgrades. A widely used method for this computation involves running a large number of DPV deployment scenarios at specific worst-case time points of the year, such as combinations of maximum and minimum loading and DPV generation as was done in [2] and [3] or just the maximum DPV-to-load ratio [4]. We refer to these studies as 'static' or 'snapshot' hosting capacity studies.

However, analysis of only these time points might provide a conservative estimate because these conditions might exist for a small percentage of the year, and these instantaneous violations, if short lived, might not count as violations per existing standards. Additionally, these types of studies cannot faithfully capture the control schemes and time delays of voltage control devices. Wang et al. proposed a robust

optimization method for optimal operation of on-load tap changers and static Var compensators but used a static hosting capacity study to evaluate the approach [5]. Thus potentially more harmful system states might not have been captured, such as sudden load or solar changes before regulation equipment is able to change taps because of their time delay settings, leading to voltage spikes or sags. Inaccurate determination of hosting capacity can trigger delays from unnecessary engineering analysis or deter solar installations in areas that are actually suitable.

In addition to these potential issues with snapshot or static DPV hosting capacity studies, there are other parameters that cannot be determined using these studies. Recent revisions to interconnection standards such as IEEE 1547-2018 [6], Rule 21 [7] and Rule 14H [8] have mandated the use of smart inverters with reactive power priority mode enabled. Smart inverter functions such as volt-var while working in reactive power priority mode might lead to curtailment of real power generation [9]. Secondly, changes in power flow in the feeder because of injections from DPV units will impact the total losses. Both of these

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can have tangible economic impacts on the end consumers and the utilities, but they cannot be quantified in a static study. Accurate assessment of hosting capacity enhancement using network reconfiguration [10] or centralized [11] and decentralized DPV control schemes also requires time-series analysis.

To address these issues, some time-varying simulations have been recently introduced into DPV hosting capacity studies, such as the use of two representative 24-hour periods [12,13]; however, these efforts use long (e.g., hourly) time steps that might not accurately capture cloud-induced variability or corresponding voltage control device responses, which typically require 1-minute or faster analysis for lower errors [14]. Dubey et al. highlighted the importance of yearly simulations for evaluating hosting capacity [15] but used hourly time steps and focused only on voltages using static ANSI C84.1-2016 range A [16] as thresholds, even though the ANSI and EN50160 [17] standards recommend using a moving average voltage value.

This paper addresses these issues by proposing a new dynamic DPV hosting capacity method and corresponding metrics. This approach is based on quasi-static time-series (QSTS) simulation, one of four special system impact studies described in a draft of the IEEE guide [18] on conducting DER distribution impact studies [14]. Per the guide, quasi-static simulation “refers to a sequence of steady-state power flow, where discrete controls, such as capacitor switch controllers, transformer tap changers, automatic switches, and relays, may change their state from one step to the next”. This study has previously been used for conducting impact studies of control schemes for smart inverters [19] and voltage regulating devices [20].

The term dynamic hosting capacity has seen some use recently such as to test stability of microgrids during switching events using dynamic equipment models [21] or to capture rapid voltage fluctuations using the worst case conditions of cloud passing over at noon of a critical day, which is likely to give conservative estimates [22]. Oliveira et al. have used this term to refer to time dependent hosting capacity while considering harmonic voltage distortion and voltage rise [23]. Thus, while some have used this term to refer to sub-second dynamics of the power system others use it to refer to duration and severity based metrics like in this paper. However, the approach proposed in this paper with respect to the latter is much more comprehensive. Also, these studies have neither used probabilistic DPV scenarios nor detailed feeder models which are considered essential for hosting capacity analysis as concluded in [24], where a review of more than 140 recent technical papers on hosting capacity analysis is presented.

To best capture the grid impacts and to determine the DPV penetration level when the dynamic hosting capacity values are reached, a new set of metrics are required [25]. These metrics should conform closely with the requirements specified in standards and not rely solely on the maximum or minimum parameter values as in static studies. By considering only the instantaneous values, the controller dynamics, and their ability to rectify violations would not be adequately captured. It is difficult to provide a single set of metric parameters and thresholds that can be universally adopted because these depend on the comfort levels of utilities in allowing small voltage deviations and overloads of their equipment. However, the set of metrics presented in Section 2 and representative thresholds presented in section 3 provide a framework for more sophisticated and accurate understanding of the hosting capacity of distribution feeders while capturing most of the potential adverse grid impacts from DPV integration during an entire year at fine time resolutions (seconds to minutes). Divshali et al. have proposed a moving average voltage metric based on EN50160 [17] standard but has presented results for only a smaller 24-hour period and does not present any thermal metrics [22]. Capturing both voltages and thermal violations is critical as highlighted in several studies [26] and [27], both of which are adequately captured in the set of metrics proposed in this paper.

A detailed network study is considered essential for any hosting capacity study [24]. A case study is thus presented in Section 3 that

compares the results obtained from static and dynamic hosting capacity studies on the model of a real feeder with probabilistic DPV scenarios generated using the widely used Monte Carlo technique [28]. The number of DPV scenarios has been reduced using a novel spatial scenario-based approach for which a mathematical formulation is provided in the appendix. Ebe et al. proposed a similar approach [29] but did not provide mathematical reasoning and applied uniform scaling factors to all DPV systems at the same nodal distance from the substation which may not be realistic as residential DPV sizes can vary widely. The PV sizes in this case study are chosen randomly from the California Solar Statistics Database, which has more than a million DPV interconnection entries [30].

The results show that dynamic hosting capacity studies can provide more accurate results, whereas static studies can be conservative. Results for parameters such as changes in system losses and DPV energy curtailment, which cannot be determined using static studies, are presented. A sensitivity analysis of the thresholds used with various metrics is also provided to help highlight why the thresholds used in this paper are good starting points and provide a basis for utilities to choose the appropriate thresholds. Thus, the contributions of this paper are twofold:

- First, a new set of dynamic hosting capacity metrics are presented that complies closely with the requirements defined in the interconnection standards and more accurately captures the grid impacts of DPV systems.
- Second, the presented case study with derived spatial Monte Carlo DPV scenarios shows that dynamic hosting capacity studies can help better capture the impact of localized DPV control schemes (such as autonomous volt-VAR) and parameters such as DPV energy curtailment, which cannot be captured in a static study.

2. Dynamic DPV hosting capacity metrics

Traditional static DPV hosting capacity studies cannot completely capture the requirements defined in interconnection standards. For instance, static hosting capacity studies usually use instantaneous limits based on ANSI C84.1-2016, but this standard does not specify any instantaneous limits [16]. The two service voltage ranges and their limits specified in this standard are applicable only to 10-minute moving voltage averages, but because static hosting capacity studies do not capture any information about how long a violation could last, these thresholds are applied to instantaneous values. This standard defines two service voltage ranges:

Range A Service Voltage (0.95 –1.050 per unit (p.u.)): “Electric supply systems shall operate such that most service voltages will be within these limits. The occurrence of service voltages outside of these limits should be infrequent” [16].

Range B Service Voltage (0.90 p.u.–1.0583 p.u.): “Although such conditions (voltages outside Range B) are a part of practical operations, they shall be limited in extent, frequency, and duration. When they occur, corrective measures shall be undertaken within a reasonable time to improve voltages to meet Range A requirements.” [16]

Through QSTS simulations, both the duration and severity of violations can be determined while correctly simulating the voltage-regulation equipment dynamics. To adequately use this information while being consistent with the standards, new metrics are required. The set of metrics proposed here looks at the various aspects of these definitions, such as considering moving voltage averages while evaluating violations and allowing minor infrequent violations. Multiple metrics needed to be developed to adequately assess all aspects of these definitions and commonly used industry practices. All of these metrics were given equal consideration while determining the dynamic hosting capacity of the feeder.

Metrics 1–4 capture ANSI Range A violations in the feeder. These four metrics also provide reasonable headroom for which a node (represents phases of a feeder bus) can be outside ANSI Range A bounds

Table 1
Summary of proposed dynamic hosting capacity metrics.

Metric Characteristics		Metric	Violation evaluated
Voltage Metrics	ANSI Range A Metrics	Duration between ANSI A and B Ranges (M1)	Total time the feeder had only ANSI Range A violations
		Maximum duration of violation per node (M2)	Total time each node experienced ANSI Range A violations
		n-minute moving voltage average (M3)	Total time the feeder's 5-minute moving voltage average was outside ANSI Range A limits
		Percentage customer hours of violations (M4)	(Percentage of customers) x (hours of ANSI Range A violations)
	ANSI Range B Metrics	Severity of nodal violations (M5)	Maximum voltage observed on each node
		Number of ANSI Range B violations (M6)	Number of time points per year with ANSI Range B violations
Thermal Metrics	Transformers	Instantaneous transformer thermal violations (M7.1)	Maximum thermal loading observed on any transformer over the year
		n-hour moving transformer loading aver-age (M7.2)	2-hour moving transformer loading average should not exceed 120% of rated kVA
	Lines	Instantaneous line thermal violations (M8.1)	Maximum thermal loading observed on any line over the year
		n-hour moving line loading aver-age (M8.2)	1-hour moving line loading average should not exceed 100% of rated kVA

Table 2
Commonly used abbreviations.

Symbol	Description
T	Total simulation duration (1 year = 525,600 min)
V_{BU}	ANSI C84.1-2016 Range B upper limit
V_{BL}	ANSI C84.1-2016 Range B lower limit
V_{AU}	ANSI C84.1-2016 Range A upper limit
V_{AL}	ANSI C84.1-2016 Range A lower limit
B	Set of all primary and secondary buses in the feeder
N	Number of buses in the feeder
ΔT	Simulation time step (1-minute)
V_{b_i}	Voltage on any bus $b \in B$, $i \in N$
s	Any simulated scenario
S_{xfmr}	Set of all distribution transformers in the feeder
M	Length of set S_{xfmr}
C	Set of all overhead or underground conductors or cables in the feeder
P	Length of set C

provided that the violations are corrected in a timely manner, consistent with the requirements of the standard. Metrics 5 and 6 capture ANSI Range B violations in the feeders. These metrics capture both the severity of Range B violations on each node and the total number of time points in the study duration when these violations occurred. Metrics 7 and 8 capture the transformer and line thermal loading violations, respectively. The thresholds used for these metrics in the case study are presented in Section 3. However, these metrics can have any threshold value based on the comfort levels of utilities, which makes the dynamic hosting capacity framework robust and extendable. These metrics are summarized in Table 1, and a more detailed discussion follows.

Capacitor bank, in-line voltage regulator and substation load tap changer operations were recorded but were not included as a limiting metric because more research is required to determine their implications on device lifetime and associated operation-and-maintenance costs. It was difficult to find standards that give accurate thresholds for all of these metrics. So based on discussions with utility partners and regulatory agencies from the US, Europe and Australia, and the experience of distribution system experts, thresholds and headroom were chosen whenever relevant literature for parameters could not be found. The dynamic DPV hosting capacity of a metric was reached when the metric values for any DPV scenario exceeded its threshold. A list of abbreviations commonly used in this and the subsequent sections is provided in Table 2.

2.1. Metric 1: Duration between ANSI A and B Ranges

This metric determines the total time points when any instantaneous nodal voltage in the feeder violates ANSI Range A voltage limits but all nodes in the feeder are within ANSI Range B limits. This metric gives an idea about the severity of the nodal violations. So if the violations are all within ANSI Range B bounds, then a small amount of headroom can be provided to allow violations to be outside Range A bounds for a short duration. The total time of the violations (T_{viol}^{M1}) was determined using Eqs. (1)–(4).

$$T_{viol}^{M1} = \sum_{t=0}^T T_{viol}(t), \quad (1)$$

where,

$$T_{viol}(t) = \begin{cases} 0 & \text{if } V_b^{max}(t) > V_{BU} \text{ or } V_b^{min}(t) < V_{BL} \\ \Delta T & \text{if } V_{AU} < V_b^{max}(t) < V_{BU} \text{ and } \\ & V_{AL} > V_b^{min}(t) > V_{BL} \\ 0 & \text{if } V_b^{max}(t) < V_{AU} \text{ and } V_b^{min}(t) > V_{AL} \end{cases} \quad (2)$$

$$V_b^{max}(t) = \max(V_b \forall b \in B \text{ at any time } t \in T) \quad (3)$$

$$V_b^{min}(t) = \min(V_b \forall b \in B \text{ at any time } t \in T) \quad (4)$$

The time violation was determined for all DPV scenarios $T_{viol}^{M1}(s_{DPV})$ and the base scenario without additional DPV systems $T_{viol}^{M1}(s_{base})$.

2.2. Metric 2: Maximum duration of violation per node

It is important to ensure that the violations do not impact a single node for a significant portion of the analysis duration. This would ensure that none of the customer locations are disproportionately impacted by the integration of DPV units. In this metric, the total time for which each node experienced voltages outside ANSI Range A limits is stored in a vector for the base case with no DPV $T_{viol}^{M2}(s_{base})$ and for all DPV scenarios $T_{viol}^{M2}(s_{DPV})$ as shown in Eqs. (6)–(8). The maximum time T_{viol}^{M2} from the base violations vector forms the baseline threshold for all DPV scenarios (5). Because ANSI Range A bounds are considered, a reasonable amount of headroom is also provided.

$$T_{viol}^{M2} = \max(T_{viol}^{M2}(s_{base})), \quad (5)$$

where,

$$T_{viol}^{M2}(s_{base}) = [T_{viol}(b_1), T_{viol}(b_2), \dots, T_{viol}(b_N)] \quad (6)$$

$$T_{viol}(b_i) = \sum_{t=0}^T T_{viol}^{b_i}(t), \forall i \in B \quad (7)$$

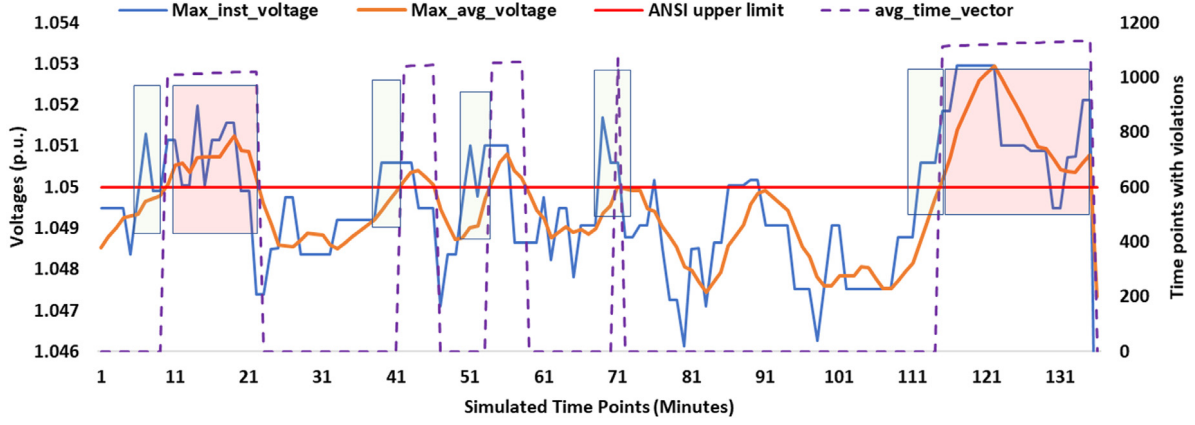


Fig. 1. Moving Average Window for Voltages; M3 allows momentary violations (green boxes) but captures the more serious sustained violations (red boxes) in a time vector (purple dotted line on secondary Y-axis).

$$T_{viol}^{b_i}(t) = \begin{cases} 0 & \text{if } V_{AL} < V_{b_i}(t) < V_{AU} \\ \Delta T & \text{if } V_{b_i}(t) > V_{AU} \text{ or } V_{b_i}(t) < V_{AL} \\ \text{at any time } t \in T \end{cases} \quad (8)$$

2.3. Metric 3: n -minute moving voltage average

Metrics M1 and M2 capture the nodal violations across the feeder during the entire year. Any time instant where ANSI Range A voltage limits are violated can contribute to the total duration; however, based on the ANSI C84.1-2016 standard and discussions with utilities, an n -minute moving voltage average window is used instead of instantaneous values to determine violations. The impact of using a moving window is illustrated in Fig. 1. This figure shows some instances where the n -minute moving average allowed momentary violations (green boxes) but captured the more serious sustained violations (red boxes) in a time vector. This metric was implemented using Eqs. (9)–(12). Once the window was filled after the first n minutes of the simulation, an n -minute moving average was calculated for each bus b at each time point t using (9) to create a vector of average voltages $\mathbf{v}_B^{n-avg}(t)$ (10).

$$V_{b_i}^{n-avg}(t) = \frac{1}{n} \sum_{k=t-n}^t V_{b_i}(k), \quad i \in N, \quad t \in T \quad (9)$$

$$\mathbf{v}_B^{n-avg}(t) = [V_{b_1}^{n-avg}(t), V_{b_2}^{n-avg}(t), \dots, V_{b_N}^{n-avg}(t)] \quad (10)$$

If the maximum or minimum value of this vector was outside ANSI Range A, then the total time violation was incremented by the simulation time step. This was repeated for all time points to get the total time of violation outside M3, T_{viol}^{M3} (11). The violation duration of M3 was determined for the base case without DPV ($T_{viol}^{M3}(s_{base})$) and all DPV scenarios ($T_{viol}^{M3}(s_{DPV})$).

$$T_{viol}^{M3} = \sum_{t=n}^T T_{viol}(t), \quad (11)$$

where,

$$T_{viol} = \begin{cases} 0 & \text{if } \max(\mathbf{v}_B^{n-avg}(t)) < V_{AU} \text{ and } \min(\mathbf{v}_B^{n-avg}(t)) > V_{AL} \\ \Delta T & \text{if } \max(\mathbf{v}_B^{n-avg}(t)) > V_{AU} \text{ or } \min(\mathbf{v}_B^{n-avg}(t)) < V_{AL} \end{cases} \quad (12)$$

2.4. Metric 4: Percentage customer hours of violations

Metrics M1–M3 look at the maximum severity of the nodal violations anywhere on the feeder and the duration of the violations on individual nodes; however, these metrics do not capture the percentage of customers experiencing nodal violations as DPV penetration

increases. Even a single node continuously experiencing violations can impact these metrics. So although these metrics ensure that no customers experience violations persistently, it is also important to ensure that a large percentage of customers do not experience violations during a major part of the year. So in this metric the percentage of nodes experiencing instantaneous voltages outside ANSI Range A limits were determined and multiplied with the simulation time step ΔT , as shown in Eqs. (13) and (14). The customer nodal violation parameter $C_{viol}^{M4}(t)$ was summed over the entire simulation duration to get $C_{cust_hr}^{M4}$, which provided an estimate of the total time for which the nodes experienced ANSI Range A violations for the base case without DPV and all DPV scenarios.

$$C_{cust_hr}^{M4} = \sum_{t=0}^T C_{viol}^{M4}(t) \quad (13)$$

where,

$$C_{viol}^{M4}(t) = \%N * \Delta T, \quad (14)$$

$$\%N = \frac{1}{N} \sum_{i=1}^N b_i, \quad (15)$$

$$b_i = \begin{cases} 0 & \text{if } V_{b_i} < V_{AU} \text{ and } V_{b_i} > V_{AL} \\ 1 & \text{if } V_{b_i} > V_{AU} \text{ or } V_{b_i} < V_{AL} \end{cases} \quad (16)$$

2.5. Metric 5: Severity of nodal violations

Metrics M1–M4 use ANSI Range A bounds and limit violations to within a small percentage of the simulation duration or the baseline value; however, a violation outside ANSI A bounds could also be outside ANSI B bounds, which is not an acceptable operating condition. In this metric, the instantaneous maximum (and minimum) voltages observed at each node during the simulation are stored in vectors for the base case without DPV and all DPV scenarios, as shown in Eqs. (17)–(19). Voltages less than ANSI Range B limits were not observed, so results for minimum voltages are not shown; however, the same set of equations can be used to assess the severity of undervoltage violations as well.

$$\mathbf{v}_{max}(s_{base}) = [v_{b_1}^{max}(s_{base}), v_{b_2}^{max}(s_{base}), \dots, v_{b_N}^{max}(s_{base})] \quad (17)$$

$$\mathbf{v}_{max}(s_{DPV}) = [v_{b_1}^{max}(s_{DPV}), v_{b_2}^{max}(s_{DPV}), \dots, v_{b_N}^{max}(s_{DPV})] \quad (18)$$

where,

$$v_{b_i}^{max} = \max(v_{b_i}(t), \forall t \in T) \quad (19)$$

The thresholds for this metric can be different for each node based on the voltages observed in the base scenario without DPV. The threshold used is the ANSI Range B upper limit, but if ANSI Range B is exceeded in the baseline, then the baseline value is used as the threshold.

2.6. Metric 6: Number of ANSI Range B violations

While metric 5 limits the severity of the nodal violations, metric 6 restricts the number of time points experiencing ANSI Range B violations to be no more than the baseline ($N_{viol}^{M6}(s_{base})$) with a small margin (T_{th}^{M6}). This metric ensures capturing the DPV scenarios where a node experiences less severe Range B violations than in the baseline but more often, as shown in Eqs. (20)–(21).

$$N_{viol}^{M6}(s_{DPV}) = \sum_{t=0}^T n_{viol}^{M6}(t), \quad (20)$$

where,

$$n_{viol}^{M6}(t) = \begin{cases} 0 & \text{if } V_b^{max}(t) < V_{BU} \text{ and } V_b^{min}(t) > V_{BL} \\ 1 & \text{if } V_b^{max}(t) > V_{BU} \text{ or } V_b^{min}(t) < V_{BL} \end{cases} \quad (21)$$

$V_b^{max}(t)$ and $V_b^{min}(t)$ are defined in Eqs. (3) and (4)

2.7. Metric 7: Transformer thermal violations

Metrics 7 and 8 capture equipment thermal loading by comparing the current flowing per phase through the device terminals with their per-phase ampere rating. Both instantaneous and moving average thermal loading are considered.

2.7.1. Metric 7.1: Instantaneous transformer thermal violations

In this metric, the maximum current flowing through any phase of a transformer $I_{T_i}^{max}$ is compared with its per-phase ampere rating $I_{T_i}^{lim}$ to get its percentage loading as shown in Eqs. (22)–(25). The percentage loading of any transformer at any time point, $L_{T_i}^{max}$ is not allowed to exceed a user-specified threshold, $T_{th}^{M7.1}$.

$$L_{T_i}^{max} = \max(I_{T_i}^{max}), \quad (22)$$

where,

$$I_{T_i}^{max} = [L_{T_1}^{max}, L_{T_2}^{max}, \dots, L_{T_M}^{max}] \quad (23)$$

$$L_{T_i}^{max} = \max(L_{T_i}(t), \forall t \in T \text{ and for any } i \in M) \quad (24)$$

$$L_{T_i}(t) = I_{T_i}^{max} / I_{T_i}^{lim} \quad (25)$$

2.7.2. Metric 7.2: n-hour moving transformer loading average

Metric 7.1 ensures that transformers are not severely overloaded at any time point. It is also important to ensure that transformers are not overloaded for a long duration. In this metric, similar to M3, an n-hour moving average window is used on each transformer (26) and this average loading is stored in a vector (27). The maximum average loading from this vector is not allowed to exceed a user-specified threshold, $T_{th}^{M7.2}$.

$$L_{T_i}^{n-avg}(t) = \frac{1}{n} \sum_{k=t-n}^t L_{T_i}(k), \quad i \in M, t \in T \quad (26)$$

$L_{T_i}(t)$ is given in equation (31)

$$L_{T_i}^{n-avg}(t) = [L_{T_1}^{n-avg}(t), L_{T_2}^{n-avg}(t), \dots, L_{T_M}^{n-avg}(t)] \quad (27)$$

2.8. Metric 8: Line thermal violations

The same metrics used to determine the transformer thermal violations are used to determine line thermal violations as well. All lines belonging to set C of length P , as described in Table 2, are considered; however, the thresholds used for line loading metrics can be different from transformer loading metrics.

2.9. Determining dynamic DPV hosting capacity

Metrics M1–M8 capture the duration and severity of nodal voltages and equipment loading based on available standards and industry accepted thresholds. As discussed previously in this section, different metrics capture different aspects of a violation. For instance, a metric might aim to limit the severity of violations but not the duration, whereas another might capture the duration but not the severity. Thus, equal consideration is given to each metric when determining the dynamic DPV hosting capacity limit of a feeder, although weights may be added if required.

Algorithm 1 gives the logic used to determine the minimum dynamic DPV hosting capacity limit for a feeder using Eqs. (1)–(27). Here, the metric values observed in each DPV deployment scenario, s_{DPV} are compared against their threshold. The DPV penetration as a percentage of customers deployed with DPV systems ($\%C(s_{DPV})$) and the percentage of peak feeder loading ($\%P(s_{DPV})$) for the first scenario where a metric value is exceeded is the dynamic DPV hosting capacity limit for that particular metric. The minimum of all these individual limits gives the minimum dynamic DPV hosting capacity limit of the feeder, $D_{HC_feeder}^{min}$.

3. Case study

The proposed dynamic DPV hosting capacity study methodology was implemented on the model of a real distribution feeder. The feeder model consists of 943 nodes, including both the primary and secondary nodes, has a peak load of 1.92 MW, and operates at 4 kV. The feeder has two capacitor banks and a substation load tap changer (LTC). The feeder topology is shown in Fig. 2. The feeder is from California, a state that has already experienced significant solar penetration, because of which time series loading and solar irradiance data at finer time resolutions were available. The feeder model was created in OpenDSS, and power flow solutions at each time step were obtained using OpenDSSDirect.py, which is a cross-platform Python package that implements a direct library interface to OpenDSS [31,32].

Getting accurate time-series data for dynamic hosting capacity studies is becoming easier nowadays because of the increasing adoption of supervisory control and data acquisition (SCADA) systems and publicly available solar irradiance time-series data [33]. For the test feeder used here, 1-minute resolution SCADA loading data were available for the feeder head. This load profile was normalized and applied to all the feeder loads. Similarly, 1-minute resolution solar irradiance profile in kW/m² of the feeder was used uniformly for all the DPV systems. The hosting capacity results can be significantly different, with unique profiles for all loads and DPV systems; however, use of identical profiles should not impact the comparisons shown here because identical parameters were used with both static and dynamic hosting capacity studies. For instance, the maximum DPV-to-load ratio time point used in the static study was taken from these profiles.

3.1. Metric thresholds used

Because metrics M1–M4 described in Section 2 are based on ANSI Range A bounds, headroom was provided to allow a small percentage of violations to exist before the dynamic hosting capacity limits are violated. A conservative margin of 1% year (87.6 h) was used, which allows ANSI Range A violations to exist for 1% of all time points simulated. At the same time, the metric thresholds should be able to consider any violations that might exist in the base case with no DPV scenario and provide a small margin (10% used here) over these violations. This helps distinguish violations caused by DPV systems from the base violations and allows a small increase over the base violations before the dynamic hosting capacity of the metric is reached. For metric M3, the most conservative window size used by a utility was 5 minutes, and this is what was used in the case study instead of

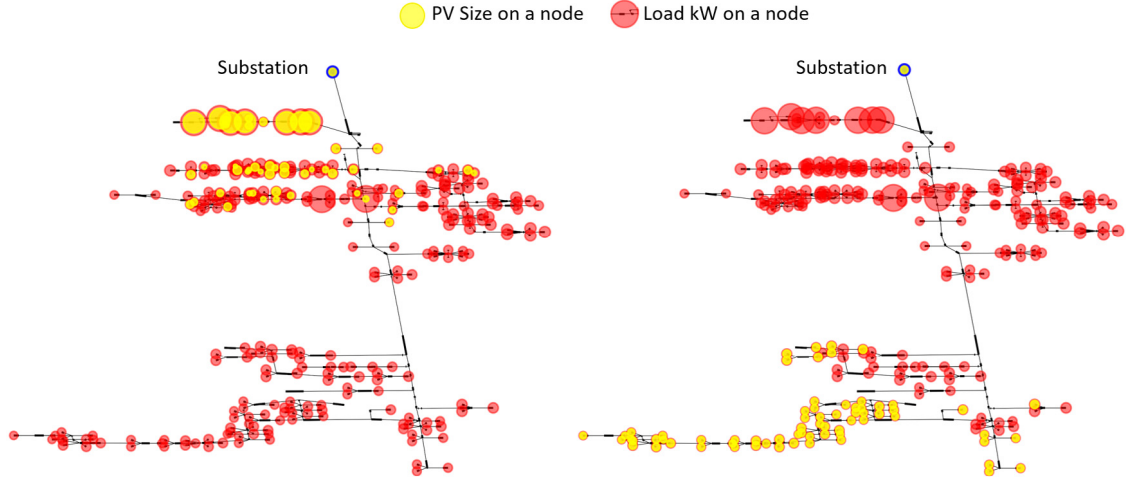


Fig. 2. Close to substation DPV locations (left) and far from substation DPV locations (right) for 15% of peak feeder load scenarios.

the 10 min suggested in the ANSI C84.1-2016 standard. Considering these points, the thresholds used for metrics M1–M4 are given by Eqs. (28)–(31).

$$T_{th}^{M1} = \max(T_{viol}^{M1}(s_{base}) * 1.1, 87.6 \text{ h}) \quad (28)$$

$$T_{th}^{M2} = \max(T_{viol}^{M2}(s_{base}) * 1.1, 87.6 \text{ h}) \quad (29)$$

$$T_{th}^{M3} = T_{viol}^{M3}(s_{base}) + 87.6 \text{ h} \quad (30)$$

$$T_{th}^{M4} = \max(C_{cust_hr}^{M4}(s_{base}) * 1.1, 87.6 \% \text{ node h}) \quad (31)$$

The thresholds for metric M5 can be different for each node based on the voltages observed in the base scenario with no DPV. So, in this case study the threshold used was the ANSI Range B upper limit, but if ANSI Range B was exceeded in the baseline, then the baseline value was used as the threshold. So, the threshold for any node b_i is given by (32).

$$T_{th}^{M5}(b_i) = \max(V_{BU}, v_{b_i}^{max}(s_{base})) \quad (32)$$

The threshold used for metric 6 allowed a small 10% margin over the baseline maximum violations as shown in Eq. (33).

$$T_{th}^{M6} = N_{viol}^{M6}(s_{base}) * 1.1 \quad (33)$$

For metric 7.1, the threshold used was taken from [34], which suggests that a transformer may be overloaded by 50% for up to 15 min provided its previous continuous loading was about 50%. So the threshold used in the case study, $T_{th}^{M7.1}$, was 150%, and equipment loading was checked once every 15 min.

$$T_{th}^{M7.1} = 150\% \quad (34)$$

For metric 7.2, a 2-hour moving transformer loading average was used and was not allowed to exceed 120% based on [34], which suggests that an oil-filled transformer may be overloaded by 10%–20% for 1–3 h provided the previous continuous loading levels were 90%–50%, respectively.

$$T_{th}^{M7.2} = 120\% \quad (35)$$

The thresholds used for metrics 8.1 and 8.2 are shown in Eqs. (36)–(37) and were determined based on discussions with utility partners.

$$T_{th}^{M8.1} = 120\% \quad (36)$$

$$T_{th}^{M8.2} = 100\% \quad (37)$$

3.2. DPV Scenario generation

The usual approach for generating DPV scenarios to perform a hosting capacity analysis involves the generation of hundreds of DPV

scenarios at each customer penetration level. All of these scenarios will have the same number of customers deployed with DPV units, but the customers selected in each scenario will be different. To limit the number of scenarios to be simulated, however, boundary scenarios may be used instead of using all random deployments [35]. The boundary scenarios used here were created by deploying DPV units close to the substation and far from substation customer locations.

The close and far scenarios are considered to be boundary scenarios because they are expected to give maximum and minimum hosting capacity values based on their reactance-to-resistance (X/R) ratio, as derived in Appendix A. Based on the results shown there, the close to substation DPV scenarios should show the highest hosting capacity values for the voltage metrics, followed by the random and far scenarios. This is because even for higher DPV injections, the close to substation buses will experience less voltage increase because of their higher X/R ratios and would be able to accommodate much higher DPV penetration levels before violations occur.

Fig. 2 shows the customer nodes deployed with DPV systems at 30% customer penetration for the close and far from substation scenarios. The 30% customer penetration scenario was chosen as it corresponds to 15% of peak feeder load for the far scenarios and is the first DPV scenario to experience voltage violations, as shown in Section 3.3.

For the case study presented here, 10 deployments were generated each for close, far, and random scenarios. Within each deployment, DPV penetration levels were increased from 0% to 200% customer penetration in a 10% step size. A level of 200% customer penetration implies that the same customer has been deployed with a larger DPV system equivalent to two smaller DPV systems. The same penetration level in the 10 deployments had the exact same number of customer nodes deployed with DPV units, but the actual set of nodes was different. The set of nodes was determined using a Monte Carlo simulation. Based on the type of load on a node, the DPV system was characterized to be a residential or commercial system. Random sampling from probability density functions of both residential and commercial DPV maximum powers acquired from the California solar DPV statistics [30] data was used to determine the DPV unit size [3]. All DPV systems had a DC-to-AC ratio of 1.15.

The DPV systems were operated in three different control modes: PF=1, PF=-0.95, and volt-VAR. To evaluate the differences between static and dynamic DPV hosting capacity studies with commonly used localized control schemes, volt-VAR mode was operated in ‘reactive power priority’ using the IEEE 1547-2018 Category A volt-VAR curve [6], and ‘watt’ priority was used with PF=-0.95 mode. All DPV systems were operated in the same control mode during a simulation. Although the DPV systems could have been separated easily into batches with

Initialize metric limits; $D_{HC_M_i}^{min} = []$, $i \in [1, 8]$

for $s \in \text{scenarios}$ do

if $T_{viol}^{M1}(s_{DPV}) > T_{th}^{M1}$ and $D_{HC_M1}^{min} = []$ then
 $D_{HC_M1}^{min} = \%C(s_{DPV})$ and $\%P(s_{DPV})$

end

if $T_{viol}^{M2}(s_{DPV}) > T_{th}^{M2}$ and $D_{HC_M2}^{min} = []$ then
 $D_{HC_M2}^{min} = \%C(s_{DPV})$ and $\%P(s_{DPV})$

end

if $T_{viol}^{M3}(s_{DPV}) > T_{th}^{M3}$ and $D_{HC_M3}^{min} = []$ then
 $D_{HC_M3}^{min} = \%C(s_{DPV})$ and $\%P(s_{DPV})$

end

if $T_{viol}^{M4}(s_{DPV}) > T_{th}^{M4}$ and $D_{HC_M4}^{min} = []$ then
 $D_{HC_M4}^{min} = \%C(s_{DPV})$ and $\%P(s_{DPV})$

end

while $D_{HC_M5}^{min} = []$ do

for $b_i \in B$ do

if $v_{b_i}^{max}(t) > T_{th}^{M5}(b_i)$ at time $t \in T$ then
 $D_{HC_M5}^{min} = \%C(s_{DPV})$ & $\%P(s_{DPV})$

end

end

end

if $N_{viol}^{M6}(s_{DPV}) > T_{th}^{M6}$ and $D_{HC_M6}^{min} = []$ then
 $D_{HC_M6}^{min} = \%C(s_{DPV})$ and $\%P(s_{DPV})$

end

if $L_T^{max}(s_{DPV}) > T_{th}^{M7.1}$ and $D_{HC_M7.1}^{min} = []$ then
 $D_{HC_M7.1}^{min} = \%C(s_{DPV})$ and $\%P(s_{DPV})$

end

for $i \in S_{xfmr}$ do

if $L_{T_i}^{n-avg}(t) > T_{th}^{M7.2}$ and $D_{HC_M7.2}^{min} = []$ then
 $D_{HC_M7.2}^{min} = \%C(s_{DPV})$ & $\%P(s_{DPV})$

end

end

if $L_C^{max}(scen) > T_{th_M8.1}$ and $D_{HC_M8.1}^{min} = []$ then
 $D_{HC_M8.1}^{min} = \%C(s_{DPV})$ and $\%P(s_{DPV})$

end

for $i \in C$ do

if $L_{C_i}^{n-avg}(t) > T_{th_M8.2}$ and $D_{HC_M8.2}^{min} = []$ then
 $D_{HC_M8.2}^{min} = \%C(s_{DPV})$ & $\%P(s_{DPV})$

end

end

end

$D_{HC_feeder}^{min} = \min[D_{HC_M1}^{min}, \dots, D_{HC_M8.2}^{min}]$

Algorithm 1: Determining dynamic DPV hosting capacity limits

different control modes within the same simulation, this would have made it difficult to compare the impacts of different control modes.

For each control mode, all three spatial scenarios were simulated at a 1-minute resolution for a year. Thus, a total of 1,890 yearlong QSTS simulations were run at a 1-minute resolution, which is close to a billion time points. These QSTS simulations were completed within 24 h by leveraging recent algorithmic advances [36], and high-performance computing systems [37]. A relatively inexpensive 4 node machine with 36 cores on each node can be used to complete all the computationally intensive volt-var simulations (600 PV scenarios, 0%–200% PV penetration in a 10% step size with an average run time of 2.4 h per simulation or about 315 million time points) in just about 10 h. These techniques have significantly reduced computational times compared with other studies looking at time series hosting capacity studies [38]. Runtimes for this paper are not based on code that has been fully optimized for runtimes or computational performance, and that there are many

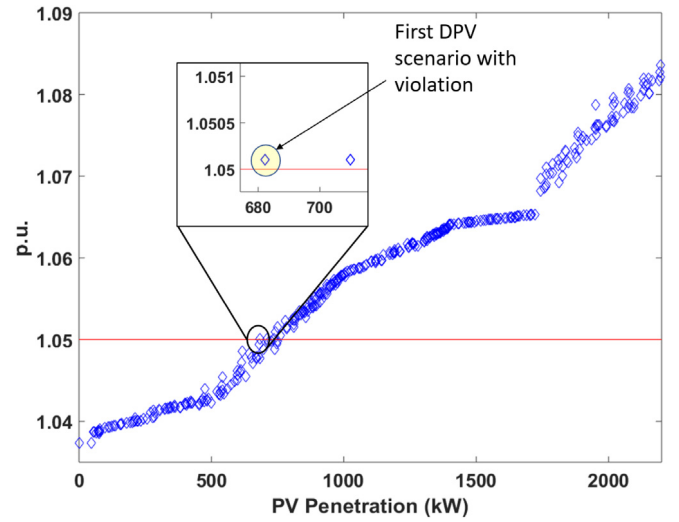


Fig. 3. Static DPV hosting capacity analysis showing maximum voltage as a function of PV capacity.

opportunities to reduce runtime, memory usage, and storage required for these methods in the future.

Of the three control modes, volt-VAR can be expected to show the maximum increase in hosting capacity values even though it uses only 25% of available reactive power because it was operated in reactive power priority. In the PF=-0.95 mode, even though the DPV systems can absorb up to 33% of available reactive power, because they were operated in watt priority and the inverters are undersized, they might not be able to provide any reactive power support during high-irradiance periods when the potential for overvoltage violations is higher, and they might behave similar to PF=1 operation. There can be deviations from these expected results, however, because of the complex topology of the test feeder compared with the circuit used in Appendix A.

3.3. DPV hosting capacity results

As described previously, three spatial scenarios were analyzed. For each spatial scenario, DPV inverters were operated in fixed power factor (PF = 1 and PF = -0.95) and volt-VAR control modes. Within each of these 9 sets, 21 DPV penetration levels with 10 different deployments for each penetration level were analyzed. The dynamic hosting capacity metrics proposed in Section 2 were evaluated for each of the 9 sets. The dynamic DPV hosting capacity limits for all 8 metrics were stored. Along with the dynamic DPV hosting capacity study, a static hosting capacity analysis was performed for each set.

3.3.1. Static hosting capacity

To conduct the static hosting capacity analysis, only the worst-case time point of maximum DPV-to-load ratio was analyzed. The following metrics were analyzed for the static study:

- Maximum voltage: The maximum of all primary and secondary bus voltages were recorded for each DPV scenario. Because the static study does not provide any information about the duration of violations, ANSI Range A limits were used as the threshold. The results obtained for close spatial scenarios for PF=1 control mode are shown in Fig. 3. The equivalent DPV kilowatt when the first violation was recorded was about 683 kW, which is equivalent to $(683/1920.3) = 35.5\%$ of the peak feeder load. As can be seen in this figure, beyond a high DPV penetration level of 1700 kW, voltage rise is more rapid and highlights the importance of conducting detailed network studies to accurately assess DPV impacts.

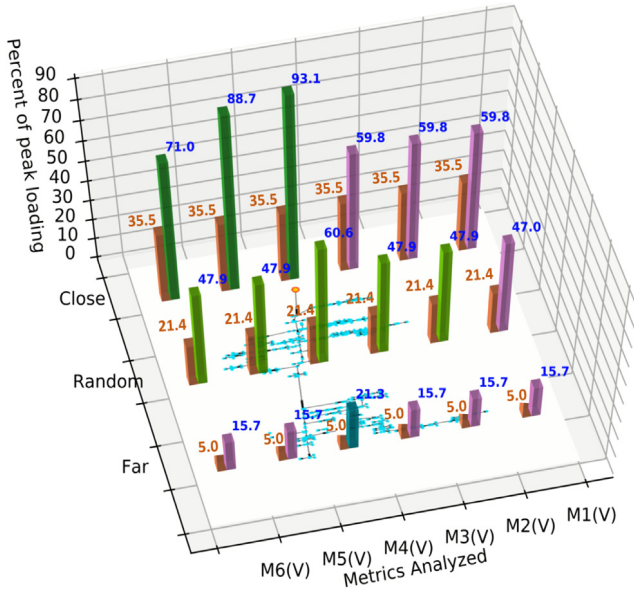


Fig. 4. Comparison of Voltage metrics with PF=1 control mode showing traditional static hosting capacity (constant orange bars) versus the suite of proposed new timeseries metrics, M1-M6, for each of the PV deployment strategies. In this scenario, the more detailed timeseries metrics reveal a noticeably higher hosting capacity for all PV deployment patterns.

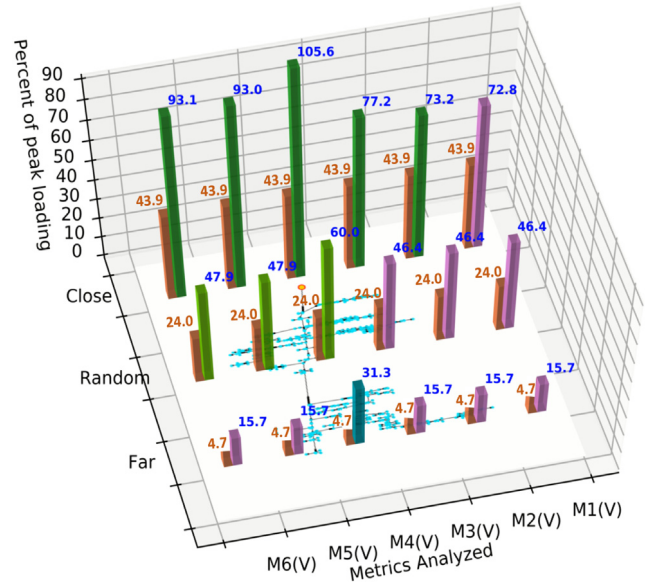


Fig. 6. Voltage metrics with volt-VAR control mode showing traditional static hosting capacity (constant orange bars) versus the suite of proposed new timeseries metrics, M1-M6, for each of the PV deployment strategies. In this scenario, the more detailed timeseries metrics reveal a noticeably higher hosting capacity for all PV deployment patterns.

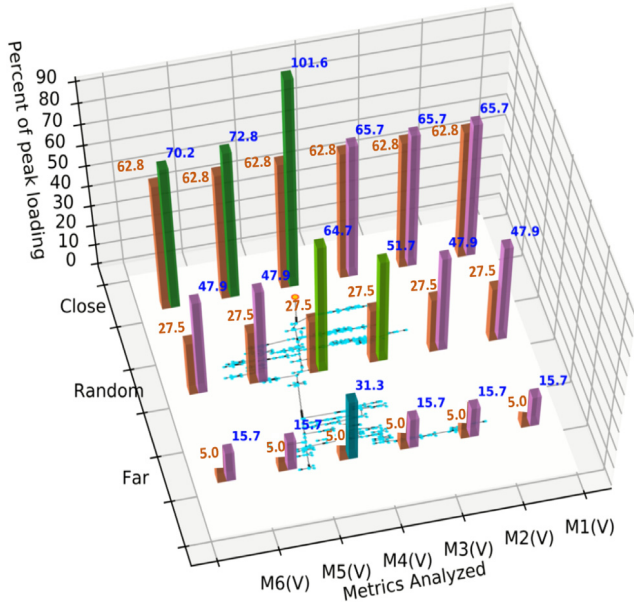


Fig. 5. Voltage metrics with PF=-0.95 (absorbing VARS) control mode showing traditional static hosting capacity (constant orange bars) versus the suite of proposed new timeseries metrics, M1-M6, for each of the PV deployment strategies. In this scenario, the increase in hosting capacity is increasingly more pronounced for PV deployment farther from the substation.

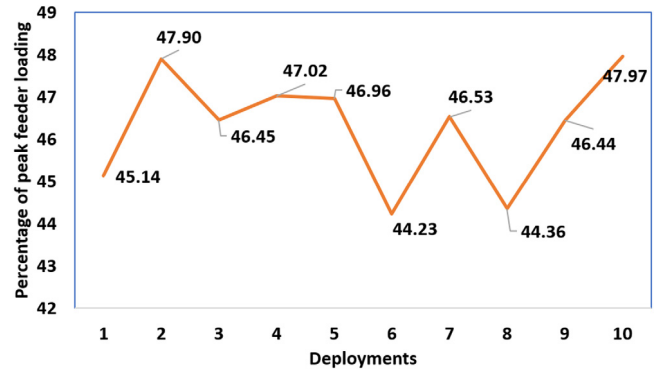


Fig. 7. DPV deployed as percentage of peak feeder loading for all 10 deployments of the 70% penetration level for random DPV scenarios.

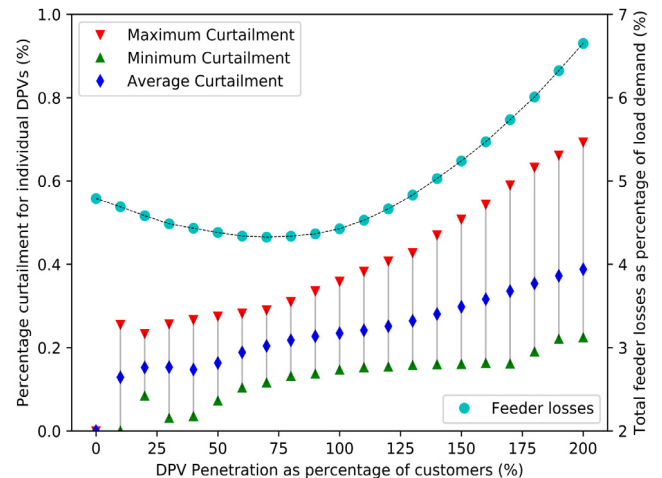


Fig. 8. Curtailment and losses for random DPV scenarios with volt-VAR control mode.

- Minimum voltage: The minimum of all primary and secondary bus voltages were recorded for each DPV scenario. As expected, no undervoltage violations were observed for any DPV scenario.
- Voltage deviations: The maximum change in voltage on each bus from the base scenario with no DPV to the DPV scenario was recorded. Any change greater than 0.03 p.u. was considered a violation.

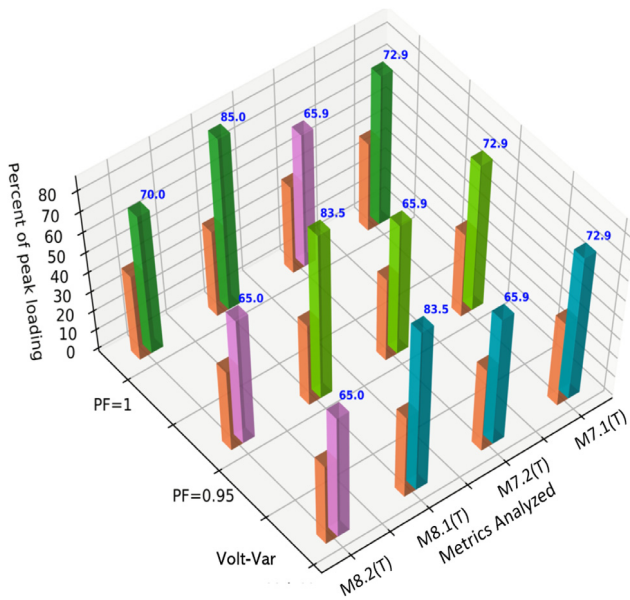


Fig. 9. Far from substation thermal violations comparing traditional static hosting capacity (constant orange bars) versus the suite of proposed new timeseries-based thermal metrics, M7–M8. In this scenario, the more detailed timeseries metrics reveal a noticeably higher hosting capacity.

- Equipment thermal violation: The maximum of all line and transformer loadings were recorded for each DPV scenario. Any loading greater than 100% was considered a violation.
- The minimum of these four static DPV hosting capacity metric limits was the static DPV hosting capacity limit of the feeder.

3.3.2. Fixed power factor results

Fig. 4 shows the dynamic DPV hosting capacity limits for the voltage metrics observed for the three spatial scenarios when all DPV systems were operated in PF = 1 control mode. The test feeder topology has also been shown to easily distinguish between the results for the spatial scenarios. The pink colored bars show the metrics that had the minimum dynamic DPV hosting value, $D_{HC_feeder}^{min}$. For reference, the feeder's static DPV hosting capacity limit has been plotted as an orange bar along with all the dynamic DPV hosting capacity limits. These results show a clear trend that the feeder can host significantly higher amounts of DPV systems at the close to substation locations, followed by random and far locations. This result is perfectly aligned with the expected trends derived in Section 3.3. The other important result is that the feeder's dynamic DPV hosting capacity limit is significantly higher than its static DPV hosting limit, by about 68%–210%. This suggests that the dynamic DPV hosting capacity study can provide more accurate estimates of the feeder's potential to host DPV systems, whereas a static study could be conservative.

Figs. 5 and 6 show plots similar to Fig. 4 for PF = -0.95 and volt-Var control modes. Even for these control modes, both static and dynamic analyses show that close to substation locations can host more DPV systems, followed by random and far locations. This suggests that DPV hosting capacity might be closely correlated with the equivalent impedances of the buses from the substation and their X/R ratios. Also, as expected, the DPV hosting capacity values are mostly higher with PF = -0.95 control mode than with the PF = 1 control mode because of the reactive power support; however, because the DPV systems were operated with 'watt' priority and the inverters were not oversized, the increase is usually not too significant. The feeder's dynamic DPV hosting capacity limit is higher than its static DPV hosting capacity limit even for this control mode.

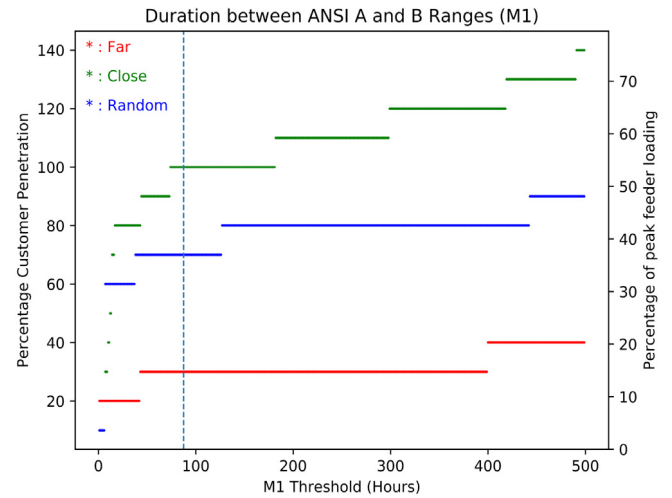


Fig. 10. Sensitivity analysis of dynamic hosting capacity metric M1.

3.3.3. Volt-Var results

The volt-Var control mode seems to have the most significant impact on increasing the DPV hosting capacity limits for close scenarios, as shown in Fig. 6. This increase was because the DPV systems were operating in reactive power priority even though the implemented IEEE 1547-2018 Category A curve absorbs only a maximum of 25% of available reactive power for voltage support and that too when the point of common coupling voltage reaches 1.1 p.u. There seems to be a reduction in dynamic hosting capacity limits for few metrics in the random scenarios when compared with the PF = 1 results; however, all of these limits are actually reached at the same customer penetration level. For instance, M2 and M3 are violated at 46.4% of peak feeder load for volt-Var, whereas they were violated at 47.9% for PF = 1, but as shown in Fig. 7, both scenarios are at the same 70% customer penetration level but different deployments. These minor differences are because the results are reported in percentage of peak load, which is easier to interpret, whereas, the DPV scenarios were generated using a 10% customer step size, and customer kilowatt values could be significantly different from one another.

3.3.4. Other impact metrics that cannot be determined in a static study

Because the DPV systems were operated with reactive power priority in volt-Var mode, there was a possibility that real power might have been curtailed during time points when both the irradiance and system voltages were higher. Thus, the total energy curtailed for each DPV system was calculated for each DPV scenario. Along with energy curtailment, the changes in the total system losses were also recorded, both of which cannot be determined in a static study. The curtailment and losses observed for the random spatial scenario with volt-Var control mode are shown in Fig. 8. The results in this figure show the maximum, minimum, and average curtailment observed for the DPV systems at each penetration level. All 10 deployments were considered at each penetration level while evaluating these results. Although energy curtailment increases with DPV penetration, the average percentage of energy curtailed is only about 0.3%, even at 200% customer DPV penetration, which is about 130% of peak feeder load. The losses show an interesting trend because initially they decrease as DPV penetration increases, but then they start increasing again. This could be because at lower DPV penetration levels load demands would have been met locally through DPV injections, thereby reducing the line flows and consequently the losses. As DPV penetration increased fur-

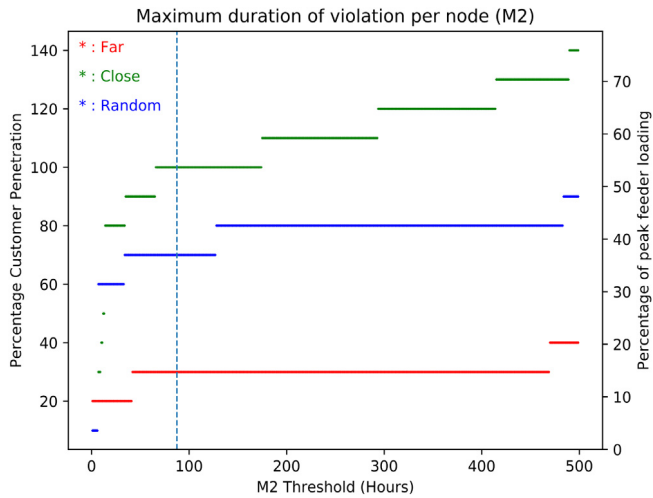


Fig. 11. Sensitivity analysis of dynamic hosting capacity metric M2.

ther, however, reverse power flows would have significantly increased the power flowing through the lines, increasing the total system losses.

3.3.5. Thermal metrics

Thermal metrics M7 and M8 were not violated for the close and random spatial scenarios until 200% customer DPV penetration; however, these metrics were violated for the far from substation spatial scenarios, although at much higher DPV penetrations than the voltage metrics. Fig. 9 shows the dynamic DPV hosting capacity limits for these metrics. This figure shows the results for the three control modes instead of the three spatial scenarios. For this spatial scenario, the equipment thermal violation metric was violated in the static DPV hosting capacity study as well, and the limit is shown as the orange bar. Even for these metrics, the figure shows that dynamic DPV hosting capacity limits were reached at much higher DPV penetration levels than with the static DPV hosting capacity limits.

3.4. Metrics sensitivity analysis

The dynamic DPV hosting capacity limits observed on the test feeder for the different DPV control modes and spatial scenarios were obtained using the threshold values proposed in Section 3.1. The sensitivity of the proposed dynamic hosting capacity voltage metrics to the threshold values was evaluated because the actual thresholds used by a utility would be dependent on their comfort level to allow relatively minor and short-term violations. Thermal metrics were not analyzed because thermal violations were observed at very high penetration levels and only for the far from substation scenarios.

Results for metrics M1 and M2 are presented here because these metrics are among the first to experience violations, as shown in Figs. 4–6. The thresholds used for these metrics in the case study were the same as described in Section 3.1 and are shown as a dotted vertical line in Figs. 10 and 11. Because the base case with no DPV had only about 7.16 h (0.08% year) of ANSI Range A violations, the 1% year margin, or 87.6 h, was used as the threshold for both M1 and M2.

For the sensitivity analysis, the thresholds were varied from 1 h to 500 h, and the corresponding dynamic DPV hosting capacity limits were recorded. The data stored from the original QSTS simulations were sufficient to complete this analysis, and no additional simulations were required. The sensitivity analysis results obtained for the two metrics for the PF = 1 control mode and the three spatial DPV scenarios are shown in Figs. 10 and 11, respectively. The plots show the DPV customer penetration level (on the primary Y-axis) when the metric was

violated for a given threshold value (on the X-axis). The secondary Y-axis shows the corresponding DPV penetration as a percentage of peak feeder loading values for reference; however, primary Y-axis values should be used because the scenarios were generated using a 10% customer penetration step, and thus the secondary Y-axis values might not be accurate for these figures.

These figures clearly show that the metrics are indeed sensitive to the threshold values, especially in the region lying to the left of the dotted line (which represents the threshold proposed in Section 3.1). In this region, even small changes in the threshold values can cause significant changes in dynamic hosting capacity value for the metrics. To the right side of this dotted line, however, a significantly larger change in threshold values is required before the dynamic hosting capacity changes. This suggests that the implemented 1% year margin is a good estimate for the threshold because it accurately captures the DPV scenario beyond which an unacceptably high number of violations will need to be allowed before the dynamic hosting capacity increases. Also, the expected dynamic hosting capacity trend of increased hosting capacity values for the close to substation scenarios followed by random and far is followed for almost the entire range of threshold values and for both metrics.

4. Conclusions

This paper presented a detailed methodology for determining the dynamic distributed photovoltaic hosting capacity study of distribution feeders. To complete a dynamic distributed photovoltaic hosting study, a set of metrics was proposed that considered the nodal voltages and equipment loading. These metrics captured both the severity and duration of the violations. They also conformed closely with the available standards, and their thresholds could be configured based on the comfort level of utilities to allow infrequent and short-term violations.

Through a case study on a real feeder model, it was shown that dynamic hosting capacity more accurately captured distributed photovoltaic impacts on the grid and when upgrades were needed to comply with existing standards. Dynamic hosting capacity was also shown to be able to capture other system effects that a static study cannot, such as energy losses in the distribution system or any curtailment from DPV resulting from the use of advanced inverter functions. A theoretical relation was derived to get expected hosting capacity trends for different distributed photovoltaic spatial scenarios and control modes. Through a sensitivity analysis, it was shown that the threshold values used in this study accurately captured the distributed photovoltaic scenario beyond which an unacceptably high number of violations would need to be allowed before the dynamic hosting capacity increased.

Future work involves implementing this methodology on a larger set of distribution feeders so that the proposed set of metrics and their thresholds can be fine-tuned. This framework will also be used to test the efficacy of distributed energy resource control algorithms, such as distributed energy resource management systems.

CRedit authorship contribution statement

Akshay Kumar Jain: Conceptualization, Formal analysis, Methodology, Visualization, Writing - original draft. **Kelsey Horowitz:** Methodology, Project administration, Funding acquisition, Investigation, Writing - review & editing. **Fei Ding:** Methodology, Software, Formal analysis, Investigation, Writing - review & editing. **Kwami Senam Sedzro:** Methodology, Software, Formal analysis, Investigation, Writing - review & editing. **Bryan Palmintier:** Conceptualization, Supervision, Writing - review & editing. **Barry Mather:** Conceptualization, Supervision, Project administration, Writing - review & editing. **Himanshu Jain:** Conceptualization, Software, Investigation.

Declaration of competing interest

No author associated with this paper has disclosed any potential or pertinent conflicts which may be perceived to have impending conflict with this work. For full disclosure statements refer to <https://doi.org/10.1016/j.apenergy.2020.115633>.

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Appendix A. Expected improvements with reactive power support

Voltages are usually the first parameter to experience violations as the amount of DPV penetration on a feeder increases. The change in voltages and the improvements achievable as a result of reactive power support are related to the (X/R) ratio of the DPV buses from the substation. A relation can be derived using a circuit model, as shown in Fig. 12. This circuit shows a substation source bus maintained at a constant voltage of V_0 p.u. supplying a DPV bus with an equivalent impedance of $R + jX$ p.u.. The voltage $V = (V_R + jV_I)$ at the DPV bus will then be a function of R and X values and the real power P being injected by the DPV system. The reactive power injection Q from the DPV system will be a function of P and the control mode. If the real power being injected is -1 p.u. (negative as power is being injected), then for inductive operation at PF = -0.95, the DPV system will absorb $Q = -1 * \tan(\cos(-0.95)) = 0.328$ p.u. reactive power (positive because reactive power is being absorbed). The relation for the DPV bus voltage V as a function of R , X , P , Q and V_0 can be derived as shown in Eqs. (38)–(47).

$$V_0 = V + I * (R + jX) \quad (38)$$

$$VI^* = (P + jQ) \quad (39)$$

$$I = \frac{(P + jQ)^*}{V^*} = \frac{(P - jQ)}{V^*} \quad (40)$$

Using (40) in (38),

$$V_0 = V + \frac{(P - jQ)}{V^*} * (R + jX) \quad (41)$$

Rearranging,

$$V_0 * (V_R - jV_I) = (V_R^2 + V_I^2) + (P - jQ) * (R + jX) \quad (42)$$

On comparing real and imaginary parts and making substitutions,

$$V_I = \frac{(-PX + QR)}{V_0} \quad (43)$$

$$V_R^2 - V_0 V_R + \frac{(PX - QR)^2}{V_0^2} + PR + QX = 0 \quad (44)$$

This quadratic equations can be solved using,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (45)$$

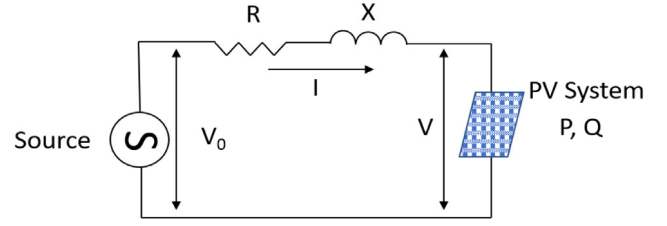


Fig. 12. Circuit showing equivalent impedance from source to a DPV bus.

where,

$$x = V_R, a = 1, b = -V_0 \quad (46)$$

$$c = RP + QX + \frac{(PX - QR)^2}{V_0^2} \quad (47)$$

These equations were used to determine the impact of the X/R ratio on the test feeder used in this case study. A short-circuit study was performed to determine the equivalent impedances of all buses from the substation. The R values in the feeder lie in the range [0.006, 2.43] Ω , X values lie in the range [0.013, 1.47] Ω , and X/R ratios lie in the range [0.441, 3.94]. The feeder operates at 4 kV, so assuming a base voltage $kV_b = 4.0$ and base apparent power $MVA_b = 10$, the base impedance will be given by (48):

$$Z_b = \frac{kV_b^2}{MVA_b} = 1.6 \Omega \quad (48)$$

To assess the impact of the X/R ratios, two values were chosen: $X/R = 2$ and 0.5 . The nearest values for any bus in the feeder were 2.056 and 0.493. For the bus with $X/R = 2.056$, the X value was 0.251 Ω , and R was 0.122 Ω . Keeping X constant, the corresponding R value for a bus with $X/R = 0.493$ would be 0.509 Ω , which is well with the range of R for this feeder. For $X/R = 2.056$, $X_{pu} = 0.157$ p.u. and $R_{pu} = 0.076$ p.u.. Similarly, for $X/R = 0.493$, $X_{pu} = 0.157$ p.u. and $R_{pu} = 0.318$ p.u.. Apart from the X and R values, the other variable that impacts V is P . Two values for P corresponding to low and high DPV penetrations were used. Because the peak feeder load is around 1.92 MW, P_{pu} of 0.0192 p.u. and 0.192 p.u., corresponding to 10% and 100% of peak feeder load were used. V_0 was kept constant at 1.04 p.u. because the substation LTC regulates it within a narrow 0.02 dead band.

Fig. 13 shows the impact of the X/R ratio and P injections on the voltage V at the DPV bus as the power factor is varied from -1 to 1 (absorbing reactive power to injecting reactive power) in a 0.05 step. The figure on the left shows that reactive power support helps reduce voltages for both high and low X/R ratios, and the improvement is much more significant for higher P injection, as expected. The magnified view on the right of Fig. 13 shows that V values are significantly higher for the bus with a lower X/R ratio than the bus with a higher X/R ratio. The impact is much more pronounced at higher power injections. This result can help explain the rationale for choosing the close, random, and far from the substation DPV scenarios, instead of a large number of random DPV scenarios, and also helps predict the expected trend in DPV hosting capacities.

Fig. 14 shows a scatter plot of bus distances from the substation and their X/R ratios for all buses in the feeder. The average distances of all buses in the 15% of peak load DPV scenario for the three spatial deployments are also plotted. The X/R ratios are in general higher for close scenarios, lowest for far scenarios, and intermediate for random. Based on the results shown in Fig. 13, the close to substation DPV scenarios should show the highest hosting capacity values for the voltage metrics, followed by the random and far scenarios. This is because even for higher DPV injections, the close to substation buses will experience less voltage increase because of their higher X/R ratios and would be able to accommodate much higher DPV penetration levels before violations occur.

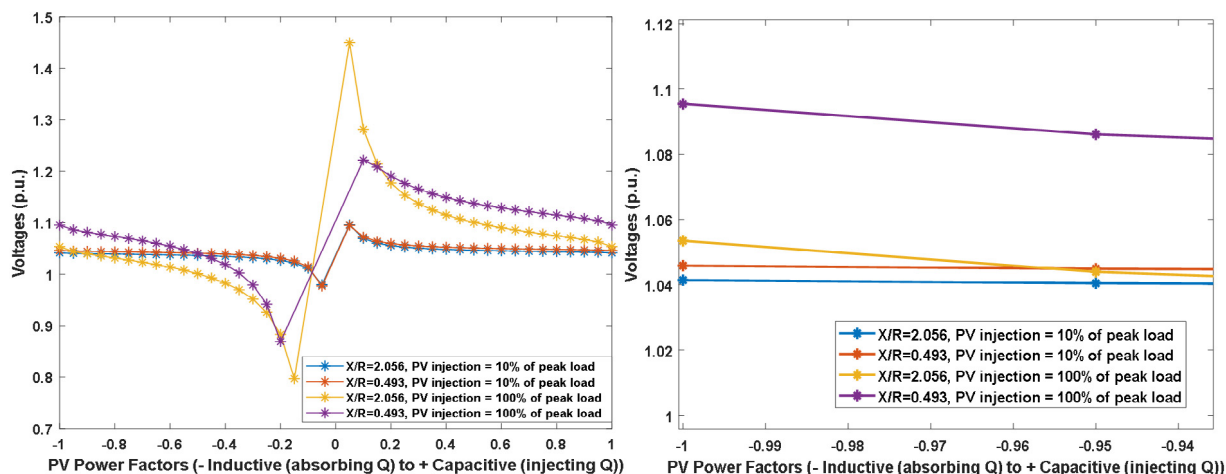


Fig. 13. Impact of X/R ratio on voltage changes from the source bus to a DPV bus in the feeder.

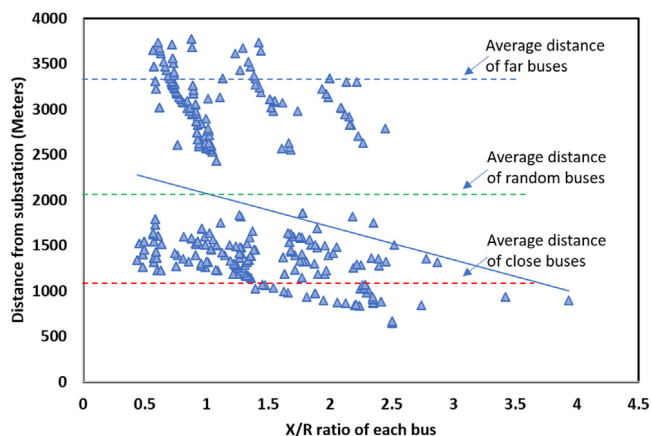


Fig. 14. XR ratios for the three spatial scenarios for 15% of peak feeder loading.

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